

1. Define Logic gate.

A logic gate is an electronic circuit with one or more inputs and a single output, which performs a basic logical function such as AND, OR, or NOT.

1. Define universal gates.

Universal gates, such as NAND and NOR gates, are gates that can perform any Boolean function without the need for any other gate type.

2. What is the primary advantage of K-map over Boolean algebra?

K-map simplifies Boolean expressions graphically, making it easier and faster to minimize functions compared to using Boolean algebra.

3. What is the function of a decoder?

A decoder is a combinational circuit that converts binary data from 'n' inputs to a maximum of 2^n unique outputs.

4. What is a multiplexer (MUX)?

A multiplexer (MUX) is a combinational circuit that selects one of many input signals and forwards it to a single output line based on selection lines.

5. What is an encoder?

An encoder is a combinational circuit that converts 2^n input lines into an n-bit output, effectively the reverse of a decoder.

6. Differentiate between combinational and sequential circuits.

In combinational circuits, the output depends only on the current inputs, while in sequential circuits, the output depends on both current inputs and previous states.

7. What is Half Adder?

A Half Adder is combinational circuit that performs 2 bit addition with two inputs and produce two outputs as Sum and Carry.

X	Y	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Sum = X Ex-OR Y and Carry = X.Y

8. What is a Flip-Flop?

A flip-flop is a basic memory element in digital electronics that can store one bit of data. It has

two stable states, representing a binary 0 and a binary 1, and is used in sequential circuits

9. What is the disadvantage that occur in SR Flip Flop?

A race condition occurs when both the Set (S) and Reset (R) inputs are simultaneously at logic 1, leading to an unstable and unpredictable output state.

10. Define Register?

A register is a group of flip-flops used to temporarily store binary data, acting as a small, fast memory within a digital system.

UNIT-II

1. What are the basic functional units of a computer?

A computer consists of five basic functional units:

- Input Unit – Takes user input (e.g., keyboard, mouse).
- Memory Unit – Stores data and instructions.
- Control Unit – Directs operations and controls other units.
- Arithmetic and Logic Unit (ALU) – Performs mathematical and logical operations.
- Output Unit – Displays or outputs the result (e.g., monitor, printer)

2. What is a Bus in a computer system? Mention the types of buses?

A bus is a communication pathway that transfers data between components of a computer. The three main types are:

Data Bus – Transfers data.

Address Bus – Carries memory addresses.

Control Bus – Sends control signals.

3. What is the difference between multiprocessors and multi-computers?

Multiprocessors have multiple processors sharing the same memory and working on a single system. Multi-computers consist of multiple computers connected in a network but working independently.

4. What is fixed-point representation?

In fixed-point representation, numbers are represented with a fixed number of digits before and after the decimal point.

5. Define floating-point representation.

Floating-point representation is a method to represent real numbers which is similar to scientific notation, where a number is expressed as a fraction (significand or mantissa) multiplied by a base (usually 2 in binary systems) raised to an exponent.

6. What is the role of artificial intelligence in the fifth generation of computers?

The fifth generation (1980-present) aims to incorporate artificial intelligence (AI), allowing computers to process natural language, solve problems autonomously, and exhibit intelligent behavior. It also focuses on parallel processing and advanced technologies like quantum computing.

7. What is the difference between fixed-point and floating-point number representation?

Fixed-point representation stores numbers as integers, with a fixed position for the decimal point, making it suitable for applications where precision is essential. Floating-point representation uses a scientific notation like format, allowing for a larger range of values but with less precision.

8. How does overflow occur in computer arithmetic?

Overflow occurs when a calculation produces a result that exceeds the range of values that can be represented within the allotted number of bits.

9. What are the types of fixed-point number representation?

Signed Magnitude – Uses the leftmost bit as the sign bit.

One's Complement – Negative numbers are represented by inverting all bits.

Two's Complement – Negative numbers are obtained by inverting bits and adding 1.

10. What is meant by overflow and underflow in floating-point arithmetic?

Overflow occurs when the result is too large for the available exponent range.

Underflow occurs when the result is too small to be represented within the exponent range.

UNIT-III

1. What is the function of the Central Processing Unit (CPU)?

The CPU is the brain of the computer, responsible for executing instructions, performing arithmetic and logical operations, and controlling data movement.

2. What are the main components of a CPU?

There are three main components of a CPU. They are

1. Arithmetic Logic Unit (ALU) – Performs arithmetic and logical operations.
2. Control Unit (CU) – Directs the execution of instructions.
3. Registers – Temporary storage for data and instructions.

3. What is the purpose of general-purpose registers in a CPU?

General-purpose registers store temporary data and intermediate results during instruction execution, reducing memory access and improving processing speed.

4. What are the basic operations of a stack?

The two primary stack operations are Push (inserting data onto the stack) and Pop (removing data from the stack). These operations help manage function calls and local variables.

5. What is an addressing mode?

Addressing modes define how the operand's location is specified within an instruction. They determine whether an operand is located in a register, memory, or provided directly as part of the instruction.

6. What is the difference between direct and indirect addressing modes?

In direct addressing, the operand's memory address is specified explicitly, while in indirect addressing, the instruction points to a memory location that contains the actual address of the operand.

7. What are the main components of an instruction format?

It consists of three fields:

Opcode Field – Specifies the operation to be performed (e.g., addition, subtraction, load, store). Operand Field(s) – Contains the source and destination operands, which can be registers, memory addresses, or immediate values.

Addressing Mode Field – Indicates how to interpret the operand (e.g., direct, indirect, immediate).

8. How does the Load instruction differ from the Store instruction?

The Load instruction transfers data from memory to a register, while the Store instruction moves data from a register to memory.

9. How does control memory work in microprogrammed control?

Control memory stores microinstructions that are fetched and decoded to generate control signals for executing machine instructions.

10. Define a control unit in a computer system.

A: The control unit is a component of the CPU that directs the flow of data and instructions within the processor. It generates control signals to coordinate the activities of other units according to the instruction being executed.